

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours**Maximum Marks:35**

Q.1 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time [15 Marks]

Q.2 Write a program to simulate Sequential (Contiguous) file allocation method. Assume disk with n number of blocks. Give value of n as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks. Write menu driven program with menu options as mentioned below and implement each option.

- a. Show Bit Vector
- b. Create New File
- c. Show Directory
- d. Delete File
- e. Exit

[15 Marks]

Q.3. Oral/Viva [5 Marks]